

I score your exam based on what you write. Demonstrate to me that you know the material. Use appropriate vocabulary. Rock on! Rock lobster!

(5 pts) 1. Is larval duration (e.g., 14 days or 60 days) a good measure of how far larvae *actually* disperse? Why?

(5 pts) 2. (a) Are two populations genetically connected or isolated if $F_{ST} = 0.46$? Are they ecologically connected or isolated? Explain.

(5 pts) (b) Is each of these two populations more likely to be sustained by larval retention or immigration? Explain.

(5 pts) 3. Why does the kelp forest distribution extend farther north on the west coast of Africa (dark shading, right) than on the east coast?



(10 pts) 4. Why is ecosystem function highest with intermittent upwelling instead of constant upwelling or downwelling?

(10 pts) 5. Why does an alternate stable state (e.g., state B) not return to its originate state (e.g., state A) when environmental conditions return to their original state? Include an explanation of hysteresis as part of your answer. You may use a diagram to aid your answer.

(3 pts E.C.) 6. Write the author(s) and title of one paper that will appear in your annotated bibliography. (Subject to verification.) Or, for 1 measly extra credit point, what plankton group is 0.02–0.2 mm in size?

(5 pts) 7. What are trait-mediated indirect interactions (TMII)?

(10 pts) 8. A tide pool in the rocky intertidal experiences high turbulence because it is exposed to waves. A nearby tide pool is protected from most waves so it experiences low to moderate turbulence. Both pools have *Enteromorpha* algae, snails, and green crabs. How will the amount of turbulence affect the snail's behavior in each pool? How will this affect the abundance of algae present in each pool?

- (20 pts) 9. Why is the phytoplankton community in the Gulf of Maine (late spring) composed mostly of microplankton while the community in the middle of the equatorial Atlantic is composed mostly of picoplankton? Your answer must explain how nutrient availability (bottom-up), competition for nutrients, and predation (top-down) contribute to phytoplankton community structure in these two regions. You may use an illustration to aid your explanation.